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NSP/004/011 Guidance on Overhead Line Clearances

1. Purpose

The purpose of this document is to specify the minimum clearances between overhead lines at all voltages up to and including 132kV and ground, general obstacles, railway and waterways property and other overhead lines.

This document has been updated to represent current practices and the specific requirements of The Electricity Safety, Quality and Continuity Regulations (ESQCR).

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NSP/004/011	Version 1 Dated July 2006	Guidance on Overhead Line Clearances
DSS/004/037	Version 1.0 - Dated 09/09/99	Guidance on Overhead Line Clearances
OHI 16	-----	NEDL – Overhead Line Clearances

2. Scope

Clearances given in this document are applicable to all new overhead lines and need not be applied retrospectively to existing lines. However if a “material alteration” is made to the line e.g. pole, conductor or crossarm change then the new clearances shall be applied.

Clearances specified in this document refer to **bare, lightly and effectively insulated line conductors** and have been determined to provide safety to the general public and protection against flashover of the line. These clearances are based on normal use of any land, buildings or structures crossed by the line, unusual situations can only be determined by local risk assessment and may require an increase in the clearances specified.

Where effectively insulated conductors are used, ground clearances over roads accessible to vehicular traffic shall be maintained as stated in Regulation 17 (2) (a) of ESQCR 2002.

Clearances over other ground or in proximity to objects may be reduced provided the effectively insulated conductor is not placed in a position where it is likely to be damaged or where persons going about normal everyday activities could come into contact with it.

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3. Overhead Line Clearances

3.1 Introduction

All clearances quoted (except those in sections 3.4, 3.6.2 & 3.9) are those listed in ENA TS 43-08 Issue 3. A direct comparison can be made to the source document ENA TS 43-08 by removing the first digit (3) from all section and table referencing.

Clearances which are bold and underlined are statutory

3.2 Siting of Overhead Lines

In siting an overhead line, special consideration should be given to the existing or future use of the land, buildings or structures over which the line is to be erected.

In such cases, a reasonable and responsible attitude must be taken towards any danger which might arise. In order to prevent danger, increased ground clearances or the use of insulated or covered conductors may be necessary, as appropriate to the situation. Guidance in the identification of high risk locations can be found by making reference to DSS/023 - Risk Assessment of Overhead Lines.

3.2 Application of Clearances

The following factors require to be taken into consideration when providing clearances to overhead lines:

- (i) Allowance shall be made for the effects of creep in conductors.
- (ii) In some cases, lines are operated at a lower voltage than that for which they are designed. It is important when specifying clearances to fixed objects that the clearances appropriate to the max potential nominal operating voltage of the line be adopted.
- (iii) When an overhead line is being erected in proximity to existing objects, the clearances shall allow for future maintenance of the object.
- (iv) When work is to be carried out, or objects are to be erected in proximity to an existing overhead line, the clearance may require to be increased substantially to allow for the operation and movement of site traffic. Detailed guidance on safe working methods are given in HSE guidance note GS6. Section 3.11 of this document has been included to allow site operators to comply with the requirements of GS6.

3.4 Allowance for Creep

Allowance must be made for the effects of creep in conductors and setting out errors as the specified clearance must be maintained for the life of the conductor.

This allowance shall be as follows:

Conductor Type	Additional Clearance Required
Copper	450mm
Aluminium Alloy	600mm
ACSR	600mm

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3.5 Proximity of Overhead Lines to High Risk Localities

An overhead line route which crosses or is in proximity to a high risk locality shall be given special consideration. Reference shall be made to DSS/023 'Risk Assessment of Overhead Lines' for guidance in the correct identification and recommended minimum mitigating measures.

Where practical all new overhead lines will be routed to avoid the need to pass in close proximity to high risk localities. However where lines are specifically identified as being in close proximity to fishing or high amenity area's, new or existing bare conductors shall be replaced with "effectively insulated conductors" eg ABC on LV lines or the use of "Lightly insulated conductors" eg XLPE covered conductors on HV lines.

3.6 Clearances to Ground and Roads and Objects

3.6.1 Clearances to Ground and Roads

The clearances specified in the table below shall not be infringed at the specified maximum conductor temperature with the conductor (including its suspension insulators if (fitted) hanging vertically in still air or deflected at any angle up to 45° from the vertical). Additional creep values detailed in clause 3.4 shall be added to these values.

Table 3.6.1 - Clearances to Ground and Roads

Item	Description	Minimum Clearance(m) System Voltage kV		
		≤33	66	132
3.6.1.1	Line conductor at any point not over a road (See note 1 & 8)	5.2	6.0	6.7
3.6.1.2	Line conductor to road surface other than as specified in item 3.6.1.3, 3.6.1.4 or 3.6.1.5 (note 2)	5.8	6.0	6.7
3.6.1.3	Line conductor to road surface of designated 6.1m "High Load Route" (note 3)	6.9	7.1	7.5
3.6.1.4	Line conductor to motorway or other road surface where "Skycradle" can be used (see note 4 & 6)	8.2	8.4	8.8
3.6.1.5	Line conductor to Motorway road surface where scaffolding is to be used on: (1) Normal 3 Lane motorways. (2) Elevated 2 Lane motorways (See note 5 & 6)	14.0	14.2	14.6
		11.0	11.2	11.6
3.6.1.6	Bare live metalwork, e.g. transformer terminals, jumper connections, etc. (See note 7)	4.3	4.3	#

Controlled Zone - Safety Rules apply.

Notes:

1. The specified ground clearance for effectively insulated conductors at locations, which are not over roads accessible to vehicular traffic, are included in Section 3.6.3. Such conductors shall be positioned so that they are not likely to be damaged or cause injury.
2. The height of any wire or cable (other than a line conductor) which is attached to a support carrying a line conductor is 5.8 m above any roadway irrespective of the line voltage. The clearances specified allow for the safe passage below the line of a vehicle of maximum height 5 m. These clearances are based on a vehicle height not exceeding 5 m (except for the 6.1 m high load routes).

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3. 'High load' routes are roads designated by the Department for Transport, for which the higher load clearance of 6.1m shall be maintained. The definition and location of roads classified as Recommended High Load Routes are laid down in Code of Practice OPS/105/017.
4. These clearances apply to situations where it is possible to use a Skycradle for conductor erection and maintenance. These clearances allow for the positioning of the Skycradle under a live circuit. Where the circuit under which the Skycradle is to be positioned and any adjacent circuit can be made dead during the slewing of the Skycradle, then these clearances can be reduced to 7.6 m for all voltages. CEEB Design Memorandum 099/100 contains guidance on the use of the Skycradle.
5. In situations where the Skycradle cannot be used to erect or maintain lines, which cross a motorway, these clearances should be adopted. They allow for the erection of scaffolding / guard netting with the overhead circuits live.
6. Should the use of Skycradles or the erection of temporary scaffolding in proximity to overhead lines be considered then appropriate guidance shall be sought relating to acceptable working methods and appropriate preparation prior to any work commencing. CEEB Design Memorandum 043/2 contains guidance on the use of scaffolding at motorways.
7. These clearances apply to supports of overhead lines that in addition support transformers, isolators, cable sealing ends, etc. These clearances do not apply to pole mounted, LV fuses as long as they are effectively insulated and the fuse carriers are in place. These clearances are not required for effectively insulated jumper connections but shall be maintained from any bare jumpers and terminals.
8. Where lines are being planned over agricultural land, and it is known that large machinery is utilised in these locations, an additional 0.3m clearance shall be added to item 3.6.1.1. in Table 3.6.1.

3.6.2 Clearance to Objects

The clearances specified in the table below shall not be infringed at the specified maximum conductor temperature with the conductor (including its suspension insulators if fitted) hanging vertically in still air or deflected at any angle up to 45° from the vertical towards the object unless otherwise specified. The clearances apply in any direction.

When an overhead line is to be erected in proximity to existing obstacles or objects the clearances should allow for future maintenance of the obstacle or object.

Note

Ground clearances and clearances to buildings for effectively insulated low voltage lines attached to poles (eg. LV ABC) are detailed separately in clauses 3.6.3.1, 3.6.3.2 and 3.6.3.3.

Table 3.6.2 - Clearance to Objects

Item	Description of Clearance	Nominal System Voltage (kV) Minimum Clearance (m)		
		≤33	66	132
3.6.2.1	Line conductor to any object which is normally accessible (including permanently mounted ladders and access platforms) or to any surface of a building. (See notes 1, 7, 8 and drawing Y101 L 0201)	4.3 (3.0)	4.3 (3.2)	4.3 (3.6)
3.6.2.2	Line conductor to any object to which access is not required AND on which a person cannot stand or lean a ladder (note 2)	0.8	1.0	1.4

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Item	Description of Clearance	Nominal System Voltage (kV) Minimum Clearance (m)		
		≤33	66	132
3.6.2.3	Line conductors to that part of a tree under / adjacent to the line and: (i) Unable to support ladder/climber (ii) Capable of supporting ladder/climber. (iii) Trees falling towards line with conductors hanging vertically only. (See note 3 & ENA TS 43-8 fig 2a or Drawing Y101 L 0202)	0.8* 3.0* 0.8	1.0 3.2 1.4	1.4 3.6 1.4
3.6.2.4	Line conductors to trees in orchards and Hop Gardens (See 4 & ENA TS 43-8 fig 2b or Drawing Y101 L 0202)	3.0	3.2	3.6
3.6.2.5	Line conductor to irrigators, slurry guns and high pressure hoses (note 5)	30.0	30.0	30.0
3.6.2.6	Line conductor to street lighting standards with: (i) Standard in normal upright position. (2) Standard falling towards line with conductor hanging vertically only. (3) Standard falling towards line. (See note 6 & ENA TS 43-8 fig 3 or Drawing Y101 L 0203)	1.7 1.7 0.4	1.9 1.9 0.7	2.3 2.3 0.8

Note 1 The (figures in brackets) are the minimum clearances that shall be maintained between an overhead line conductor and a normally accessible structure or surface of a building (walls, roof, windows etc.). They permit a person to stand on or against these structures but only allow for free movement of short hand held objects. These figures are taken directly from ENA TS 43-8 and relate to the clearances to buildings where we have carried out a full instrument survey to the object taking into account the conductor sag at the time of the observation and then extrapolating this reading into a new tension/sag of where the conductor will be when it is operating at it's maximum design temperature. The new conductor created from this calculation shall be used to produce a swing envelope drawing in PoleCad to ensure that the stated minimum clearance can be achieved with the conductor hanging vertically in still air or deflected at any angle up to 45° from the vertical towards the object.

Where clearances to buildings or objects comply with this minimum clearance we would not propose to divert overhead lines just to achieve the 4.3m clearances

The 4.3m clearance figure was created as a practical guide and good practice clearance figure that would normally be applied to the placement of new overhead lines. However it may also be used as a quick method for confirming that existing HV lines have adequate clearance to buildings as it can be applied without the need for a full instrument survey.

It is only when this figure is infringed that we need to carry out the full survey method to confirm the statutory minimum figures.

Note 66/132kV lines shall always be subjected to a full instrument survey for clearance enquiries followed by the production of swing envelope drawings in PoleCad as these spans can be significantly larger than those allowed for HV lines and typically employ the use of suspension insulators which allow greater conductor movement towards buildings.

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Detailed guidance on the avoidance of danger from electric lines on construction sites is contained in HSE guidance note GS6.

See section 3.6.3 for alternative min clearances when effectively insulated conductors are utilised.

Note 2 - Account should be taken of the possible movement of the object, e.g. flagpole in the wind. These clearances also apply to moving objects to which access is precluded during passage below the line. The height or position of the object should take into account any possible undulating or rocking movement of the object, e.g. a mobile crane jib travelling over uneven ground. Detailed guidance on clearances to electric lines on construction sites is contained in Engineering Recommendation G35.

Note 3 - Clearances quoted in 3.6.2.3 (i) & (ii) are minimum acceptable clearances but in practice larger clearances will be necessary to take account of growth rates of trees and of the swaying of trees/branches in the wind. Clearances quoted in 3.6.2.3 (iii) are recommended in order to protect lines from falling trees but due to wayleave considerations will not always be attainable. Detailed guidance on the avoidance of danger from electric lines in forests is contained in CE Electric UK Document – OPS/105/013 – Guidance on work in proximity to Overhead Electric Lines and Engineering Recommendation G55/1.

Note 4 - These clearances must be obtained vertically when any part of a tree is within 7.5m horizontally of a line. For Hop Gardens, the clearances apply to the strain wires forming the mesh supporting system.

Note 5 - The clearance quoted is for general guidance only. Detailed guidance on the use of irrigators, slurry guns and high pressure hoses in the vicinity of overhead lines is contained in Engineering Recommendation G45/1.

Note 6 - The clearances quoted in 3.6.2.6 assume that maintenance platforms will be positioned such that clearances quoted in 3.6.2.1 are maintained. Where effectively insulated conductors are used, clearances may be reduced to those indicated in Fig 4. The clearances quoted in 3.6.2.6 (iii) can be neglected if the location of the lighting column is such that impact by a vehicle is improbable.

Engineering Recommendation G39 contains guidance on maintenance of street lighting columns in proximity to overhead lines. Where for maintenance purposes the operative requires to work on the upper part of a lantern, within the clearances specified in 3.6.2.1, appropriate safety measures shall be taken, which shall be agreed in advance between the CE Electric UK and the lighting maintenance company or authority. The clearances quoted in 3.6.2.6 (ii) include additional clearance to allow for the erection of street lighting columns.

* LV effectively insulated conductors shall be positioned so as to prevent chafing.

Note 7 - Where telecommunications masts are constructed adjacent to an existing overhead line, these clearances shall be increased to provide a minimum lateral separation between any line conductor and the nearest point on the mast of 1.5 times the height of the mast or 15.0m whichever is the least.

Note 8 – Where wind turbines are constructed adjacent to an existing overhead line, the clearances in clause 3.6.2.1 shall be increased to provide a minimum lateral separation between any line conductor and the nearest point on the turbine of 1.5 times the height of the turbine where the max height of the turbine is defined as the upper tip in its vertical position.

LV – 20kV lines which may be erected with a reduced minimum clearance of 1 times the height of the structure provided they utilise covered conductors and have been fitted with appropriate vibration dampers.

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These additional clearances have been instigated to counter the significant risk of overhead lines being subjected to additional mechanical damage or reduced reliability caused by the wake effect created by wind turbines.

3.6.3 Clearance from effectively insulated low voltage lines attached to poles

Effectively insulated conductors for example LV Aerial Bundled Conductors (ABC) installed in accordance with DSS/004/041 "LV ABC Overhead Lines" or DSS/004/041/001 "Specification for the renovation of existing open wire networks to ABC" shall comply with the clearances in this section. However, clearances between other power lines and above railways, as detailed in sections 7 & 8 and tables 7.1 and 8.1 below, shall be met.

In addition to the above, the following of list of common conductor types also comply with the term "Effectively Insulated Conductors"

- a) PVC single insulated Phase and Neutral conductors situated on LV open wire network poles (not service spans)
- b) Double insulated conductors
- c) Hybrid CNE service conductor

Note:-

PVC single insulated phase and neutral conductors used for service spans and covered conductors are not classed as "Effectively insulated" and hence must use table 3.6.2

3.6.3.1 Ground Clearance

Clearances to roads shall still comply with paragraph 3.6.1 and table 3.6.1 above however clearances in other locations are provided in table 3.6.3.1 below.

Table 3.6.3.1 - Ground Clearance

Location	Minimum Clearance (m)
Along the line of hedgerows, fences and boundary walls etc.	4.0
Domestic driveways with an access width of 2.5m or less which is defined by gateposts, hedges or other fixed features.	4.3
Between buildings where there is no vehicular access.	3.5

3.6.3.2 Clearances to Buildings and Structures

In determining clearances the following conditions should be considered as appropriate, sags at the maximum working temperature of 75°C and an angular deflection of 30° at a working temperature of 30°C. Deflected conditions need not be considered if the span is effectively shielded from wind by the building or structure. The clearances in Table 3.6.3.2 below do not apply to mains or services attached directly to buildings.

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Table 3.6.3.2 - Clearance to Buildings and Structures

Location	Minimum Clearance (m)
Vertical clearance to any surface or structure that is accessible without access equipment (ENA TS 43-08 see Fig. 5 or Drawing Y101 L 0205).	3.0
Horizontal distance to any surface of a building or structure which is accessible without access equipment. See (ENA TS 43-08 Fig. 5 or Drawing Y101 L 0205).	1.0
Clearance to parts of a building or structure not normally accessible (see ENA TS 43-08 Fig. 5 or Drawing Y101 L 0205). See Note 1.	0.5
Clearance to free-standing apparatus such as street lighting columns, traffic signs, British Telecom poles or columns See (ENA TS 43-08 Fig. 5 or Drawing Y101 L 0205).	0.3

Notes:

1. This clearance is to prevent mechanical abrasion of the conductor. When connecting from a pole to a building it is only necessary to ensure that the attachment route avoids risk of abrasion.

3.6.3.3 Systems attached to Buildings

Effectively insulated conductors (eg. LV ABC) may be directly attached to buildings provided they are installed in accordance with CE Electric UK Specification DSS/004/041 "LV ABC Overhead Lines"

3.7 Clearances where power lines cross or are in close proximity

The following minimum clearances shall apply where power lines cross or are in close proximity to one another. In all cases the clearances shall be determined by the ultimate nominal system voltage of the upper or lower line, whichever is the greater.

Table 3.7.1 - Minimum clearances where power lines cross or are in close proximity

Item	Description of Clearance	Nominal System Voltage (kV)					
		Minimum Clearance (m)					
	Conductor or earth wire to:	0.4	11	20*	33	66	132
3.7.1.1	Lowest Line conductor or earth wire of upper line to highest line conductor of lower line (Note 1)	1.0	1.8	2.0	2.0	2.3	2.7
3.7.1.2	Lowest line conductor or earth wire of upper line to earth wire of lower line where erected. (Note 1)	0.7	1.4	1.6	1.6	2.3	2.7
3.7.1.3	Lowest line conductor or earth wire of upper line to any point on a support of the lower line on which a person may stand. (Note 2)	2.7	2.8	3.0	3.0	3.2	3.6
3.7.1.4	Support of upper line and any conductor of lower line (Note 2)	7.5	7.5	7.5	7.5	7.5	15.0

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Notes:

1. One of the following methods of determining clearances shall be adopted.

(a) With the upper conductors / earth wire hanging vertically and the lower conductors / earth wire deflected at 45° under the following conditions:

(i) Upper conductor at its specified maximum temperature coincident with lower conductor at an assumed temperature of 25°C less than its specified maximum temperature.

(ii) Lower conductor at a temperature of -5.6°C (no ice) coincident with the upper conductor at an assumed temperature of 20°C.

or alternatively

(b) With the upper conductor / earth wire hanging vertically at its specified maximum temperature and the lower conductor / earth wire deflected at any angle up to 45° at a temperature of -5.6°C (no ice). In localities where there is a high likelihood of conductor icing it may be appropriate to consider the effects of such icing.

2. Clearance shall be obtained with the conductor / earth wire at its specified maximum temperature and deflected by any angle up to 45°.

* 20kv assumed to be equivalent to 33kV values.

Table 3.7.2 - Minimum Clearances to other power lines (Protection against Induced voltages from Parallel Lines)

Where lines are routed parallel to existing power lines the following minimum clearances shall be provided between the centre line of the existing line and the proposed line in table 3.7.2

Table 3.7.2 – Parallel lines - Protection from induced voltages

Nonimal System Voltage (kV)	Minimum Clearance (m)
Wood Pole and Steel Mast Lines upto 66kV	18.0
66kV Double Circuit Tower Lines	30.0
66/132kV Double Circuit Tower Lines	45.0
275kV Double Circuit Tower Lines	60.0
400kV Double Circuit Tower Lines	76.0

In Cases were it is impractical to comply with the above, guidance shall be sought from the Overhead Systems Standards Manager

3.8 Clearances to Railway Crossings

Clearances to railways and their associated lines, buildings and yards are covered by the second schedule (General and Engineering) of the Condition Railway master Wayleave Agreement.

Table 3.8.1 lists the principal vertical clearances referred to in the above Agreement. For horizontal clearances to railway circuits (excluding traction wires) reference should be made to the Agreement.

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Table 3.8.1 - Principal Vertical Clearances to railways and Associated Structures

Item	Description of Clearance	Nominal System Voltage (kV) Minimum Clearance * m (ft)		
		≤33	66	132
3.8.1.1	Ground Level.	6.1 (20)	6.1 (20)	6.7 (22)
3.8.1.2	Ground Level at roads or yards where road mobile cranes are likely to be employed.	10.7 (35)	10.7 (35)	11.2 (37)
3.8.1.3	Rail Level (See note 1)	7.3 (24)	7.3 (24)	8.0 (26)
3.8.1.4	Buildings, gantries or other structures on which a man might stand and to traction wires (see Note 1)	3.0 (10)	3.0 (10)	3.7 (12)

* The imperial values take precedence since they are specified in the Agreement

Notes:

- The clearances specified in 3.8.1.3 and 3.8.1.4 do not incorporate any allowances for use of scaffolding or Skycradle across railway tracks / traction wires during erection / maintenance of overhead lines. Clearances of 2.75m (9 feet) and 4.9m (16 feet) are required between scaffold net or Skycradle boom and traction wires and rail respectively.

3.9 Clearance to Waterways

Clearances to waterways are not subject to a single national Agreement but are dealt with by agreement with the appropriate Authority. Statutory ground clearances shall form the basis of negotiations but additional clearances may be agreed for navigable waterways. For information, the following minimum clearances are usually stipulated by the Waterways Board in the YEDL licensed area.

Item	Description	Clearance (m)		
		0.433 kV	66kV	132kV
	Lowest conductor (line or earth)			
	Towpath level or adjacent bank			
1	River Trent	18.3	19.2	19.8
2	River Ouse	23.2	23.2	23.8
3	Aire & Calder Canal	14.3	15.2	15.9
4	Sheffield & South Yorkshire canal	14.3	15.2	15.9
5	Calder & Hebble Canal	14.3	15.2	15.9
6	Other Canals	11.3	15.2	12.8
7	Reservoirs	12.3	15.2	15.6

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3.10 Clearance to Telecommunication Lines

Whenever overhead lines cross or are in close proximity to British Telecom lines, the conditions laid down in the following documents shall be complied with:

Vertical lateral clearances to telecommunication lines are specified in Engineering Recommendation PO.5 for voltages up to and including 33kV.

Engineering Recommendation EB/BT.2 'Conditions for BT and Public Suppliers' Joint Use of Poles' specifies the clearance requirements for apparatus when poles are jointly used.

Various alternative methods of protection are available and the choice of which method to adopt depends upon the voltage of the power line, the number and size of the British Telecom circuits and the cost of providing protection.

The following table gives a guide to the types of protection available:-

Voltage	*Protection by CE Electric UK	Protection by British Telecom
Up to 11kV RMS between conductors or 6.6kV RMS between conductor & earth	XLPE Covered Conductors (i), PVC Insulated Conductor (ii) or Underground	PVC insulation, Insulated aerial cable or underground
20kV RMS between conductors or 11.9kV RMS between conductor & earth	XLPE Covered Conductors (i), PVC Insulated Conductor (ii) or Underground	PVC insulation, Insulated aerial cable or underground
33kV RMS between conductors or 19kV RMS between conductor & earth	Underground	PVC insulation or underground
66kV RMS between conductors or 38kV RMS between conductor & earth	Underground	Underground

- To be adopted only under very exceptional circumstances where it is impossible to protect British Telecom circuits by any other method.

(i) HV XLPE covered conductor manufactured in accordance with ENA TS 43-122 is approved for use upto and including 20kV.

(ii) Insulation shall be green PVC, minimum thickness 1.6mm, to BS6485 (This option will be removed when use of covered conductor is standardised across CE Electric UK).

When HV overhead lines and communication circuits run in proximity, the possibility of induced voltages in the communications circuits due to parallelism should be considered. Engineering Recommendation PO.5 deals in detail with the subject. Cases of difficulty in meeting British Telecom requirements for keeping the voltage induced in communication circuits below the prescribed maximum value should be referred to the Overhead Systems Standards Manager.

3.11 Work in Proximity to Overhead Lines

This section deals with the use of plant or vehicles in proximity to overhead lines. Where work is undertaken using ladders, scaffold, mobile platforms etc. then the clearances provided in Tables 3.6.1 and 3.6.2 shall be used unless other risk mitigation can be employed such as temporary shrouding of the

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overhead conductor. Whenever work is to be carried out in proximity to overhead lines consideration shall always be given to the possibility of making the line dead, or diverting it around the area affected.

The Health and Safety Executive provide guidance for the avoidance of danger from overhead lines in their Guidance Note GS6.

CE Electric UK will provide, preferably in writing, safety clearances and advice on safe working methods to those working in proximity to overhead lines. Where work can only be carried out safely with the line dead, this shall be the subject of precise written agreement between CE Electric UK and the site operators.

3.11.1 Horizontal Clearances – on sites where there will be no work or passage of plant under lines.

HSE Guidance Note GS6 recommends that the DNO should be contacted for advice for any work within 15 m of a line erected on steel towers and 9 m of a line erected on wood poles.

Table 3.11.1 details typical minimum values for horizontal separation of the lines and safety barriers.

Table 3.11.1 - Horizontal Distances to Safety Barriers

Voltage Type	≤33 kV Wood Pole	66kV Wood Pole	66kV Tower	132kV Wood Pole	132kV Tower
Minimum horizontal distances to safety barriers.	6.0m	6.0m	9.0m	6.0m	9.0m

Note: Site conditions will dictate whether this clearance is adequate and consideration shall be given to line parameters e.g. span length, maximum sag etc. when calculating an actual clearance.

3.11.2 Vertical Passing Clearances – on sites where vehicles will pass under the lines.

Table 3.11.2 details these passing clearances and Appendix 2 provides their derivation.

The clearances given in Table 3.11.2, Item 3.11.2.1 are for vehicles with fixed height loads travelling on un-metalled roads.

Where the load carried by vehicles is variable then the vertical passing clearance shall be increased. These clearances are given in Table 3.11.2, Item 3.11.2.2.

Table 3.11.2 - Vertical Passing Clearances

Item No.	Nominal System Voltage		≤33 kV	66kV	132kV
3.11.2.1	Passing Clearance fixed height loads	(m)	0.8	1.0	1.4
3.11.2.2	Passing Clearance Variable height loads	(m)	2.3	2.5	3.2

The above clearances shall be used to determine the maximum distance to the underside of barriers erected to prevent vehicles or plant from infringing these clearances whilst traversing the line. The height to

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the underside of the barrier shall be the minimum ground clearance of the line less the specified passing clearance in Table 3.11.2.

3.11.3 Vertical Clearance – on sites where work will be undertaken beneath the line.

Work beneath the line shall be deemed to be any work carried out within the minimum horizontal distances specified in Table 3.11.1 or the calculated distance (see note under 3.11.1) whichever is greater.

HSE Guidance Note GS6 provides recommendations for working under the line and uses two cases. “Work at ground level only (for example pipe laying)” and “Erection of buildings or structures underneath an overhead line”. In both cases the clearances are the same and these are shown in Table 3.11.2 Item 3.11.2.1.

3.11.3.1 Work at ground level only.

Where work is carried out at ground level only the passing clearance of fixed height loads is permissible, as HSE Guidance Note GS6 requires that no vehicle or item of plant shall reach beyond the safe clearance limit. Where plant has the capability to reach into the safe clearance limit it shall be fitted with a physical restraint in order to prevent such action.

HSE Guidance Note GS6 requires that all such work shall be “under the direct supervision of a responsible person”.

3.11.3.2 Work on buildings or structures underneath an overhead line

This includes all work under an overhead line and includes new construction, work on existing structures and demolition.

A horizontal physical barrier shall be erected to form a roof between the area of work and the overhead line such that the safe clearance limit cannot be infringed. The distances in Table 3.11.2, Item 3.11.2.1 shall be treated as a minimum necessary clearance and shall be used to calculate the height of the underside of the physical barrier.

Where a conductive material is used to form the barrier this shall be earthed.

The line shall be made dead if, during the erection of the physical barrier, safety clearances would be infringed.

3.12 Clearance of Unsupported Jumper loops

Item	Description	Clearance (m)				
		11kV	20kV	33kV	66kV	121kV
1	Unsupported jumper loop to any part of a support or stay.	0.3	0.45	0.5	0.7	0.9

These clearances shall apply:-

- On unearthed supports with the conductors in still air.
- On earthed supports with the jumper swung at any angle up to 30° towards the support.

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4. References

4.1 External Documentation

Reference	Title
ENA TS 43-08	Overhead Line Clearances
GS6	HSE guidance note GS6 (rev) 'Avoidance of Danger from Overhead Lines' ISBN 0 11885 6685.
SMCC 004	SMCC 004, "Notes of Guidance on Safe Working of Third Parties in Close Proximity to Live Overhead Conductors"
ESQCR	Statutory Instrument 2002 No. 2665, The Electricity Safety, Quality and Continuity Regulations 2002
G39/1	Engineering Recommendation G39/1, 'Model Code of Practice Covering Electrical Safety in the Planning, Installation, Commissioning and Maintenance of Public Lighting and Other Street Furniture'.
G45/1	Engineering Recommendation G45/1, 'Notes of Guidance for Electricity Boards on the Use of Irrigators, Slurry Guns and High Pressure Hoses in the Vicinity of Overhead'.
G55/1	Engineering Recommendation G55/1, 'Safe Tree Working in Proximity to Overhead Electric Lines'.
PO.5	Protection of Telecommunication Lines from Power Lines
PO.3	Engineering Recommendation PO.3, 'Technical Conditions Applicable under Master Wayleave Agreements between the Post Office and the Electricity Supply Industry for Wayleaves on Post Office Telecommunications Property'.
EB/BT.2	Engineering Recommendation EB/BT.2 'Conditions for BT and Public Suppliers' Joint Use of Poles'.

4.2 Internal documentation

Reference	Title
Y101L0201	Clearance to objects on which a person might stand
Y101L0202	Clearance to trees
Y101L0203	HV conductor clearance to lighting columns
Y101L0204	LV conductor clearances from lighting columns
Y101L0205	Clearances between structures and effectively insulated conductors installed on poles
DSS/004/041	LV Overhead Lines
DBD/004/038	Overhead Line Crossings & procedures
OPS/105/013	Guidance on Work in proximity to Overhead Electric lines
OPS/105/017	Guidance on high loads
GN70-25	Assessing Anti-climbing Guard Requirements for Overhead Line Supports.

4.3 Summary of Changes

- Clause 3.6.1 note 1, amended to provide additional clarification with respect to clearances to buildings.
- Clause 3.6.1 note 8, added to detail the requirements for clearances to wind turbines
- Clause 3.6.3 amended to provide additional guidance on the types of service conductor that can be classified as effectively insulated and can thus take advantage of the reduced clearance figures tabulated in clause 3.6.3.1 and 3.6.3.2.

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5. Definitions

Term	Definition
Application Factor	The distance (dependant upon working situation) which is added to the safety distance to determine working and access clearances
Basic Electrical Clearances	Clearance to earth ascribed to air insulation for each nominal system voltage. Basic electrical clearances do not include any additions for constructional tolerances, wind effects, etc.
Cable	A conductor or assembly of conductors, which are effectively insulated and incorporate an earthed metallic screen.
Controlled Zone	The inside of an enclosure efficiently protected by fencing not less than 2.4m (8ft.) in height or means necessary to meet the requirements of ESQCR 2002 regulation 11(b)
Covered Conductor	A design of conductor that can be lightly or effectively insulated and may be manufactured in accordance with ENA TS 43-122. (In ESQC guidance notes this is often referred to as BLX). Note – The type of covered conductor currently used in CE Electric UK is classified as lightly insulated.
Creep	The non - elastic stretch of a conductor. This consists of two parts - bedding down of the strands and the long-term stretch of conductor material.
Effectively Insulated Conductor	A line conductor which is insulated for continuous phase to phase or phase to earth contact and is protected, so far as is reasonably practicable, against mechanical damage or interference having regard to its accessibility. Note : The implication here is that insulated conductors (such as low voltage aerial bundled conductors but not high voltage CC conductors) may be placed such that they are ordinarily accessible, providing they are safe in the particular circumstances.
Jumper Connection	A connection at a support from a phase conductor to another conductor or to a terminal on transformers, switchgear, fusegear, line taps etc. at support.
Lightly Insulated Conductor	A line conductor which is insulated against momentary phase to phase or phase to earth contact and is considered as a bare conductor for clearance purposes. Note: This level of insulation may be designed to support the full phase to phase or phase to earth voltage (as appropriate). For example the covering on some types of HV or EHV CC overhead line conductors could be described as lightly insulated.
Line Conductor	A conductor used, or to be used, for conveying a supply of electricity. A line conductor is deemed to include a through jumper.
Normal Use of Land	The type of work or activity which is likely to occur on or over a particular piece of land or water.
Object	Any building, wall, fence, structure, stationary vehicle, tree, vegetation or similar with an elevation above ground level.
Overhead Line	Apparatus in the open air and above ground level coming within the scope of ESQCR 2002.
Safety Distance	Distance from nearest exposed conductor or from an insulator supporting a conductor is maintained to avoid danger.

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Term	Definition
Specific Maximum Temperature	The likely maximum temperature of the conductor resulting from a combination of climatic conditions and the related electrical load under normal operating conditions.
System Voltage	The nominal RMS phase to phase voltage of a three phase AC system
Wire	A wire which is not designed to convey electricity but which is attached to a support carrying line conductors, eg. Flying stay wire.

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager	G Hammel	18/02/2010

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	23/02/2010

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	18/02/2010

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	08/03/2010

6.5 Approval

Approval is given for the content of this document

		Sign	Date
Andy Bilclough	Overhead Programme Manager	Andy Bilclough	25/02/2010

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
S Caskey	Head of Programme Delivery	S Caskey	25/02/2010

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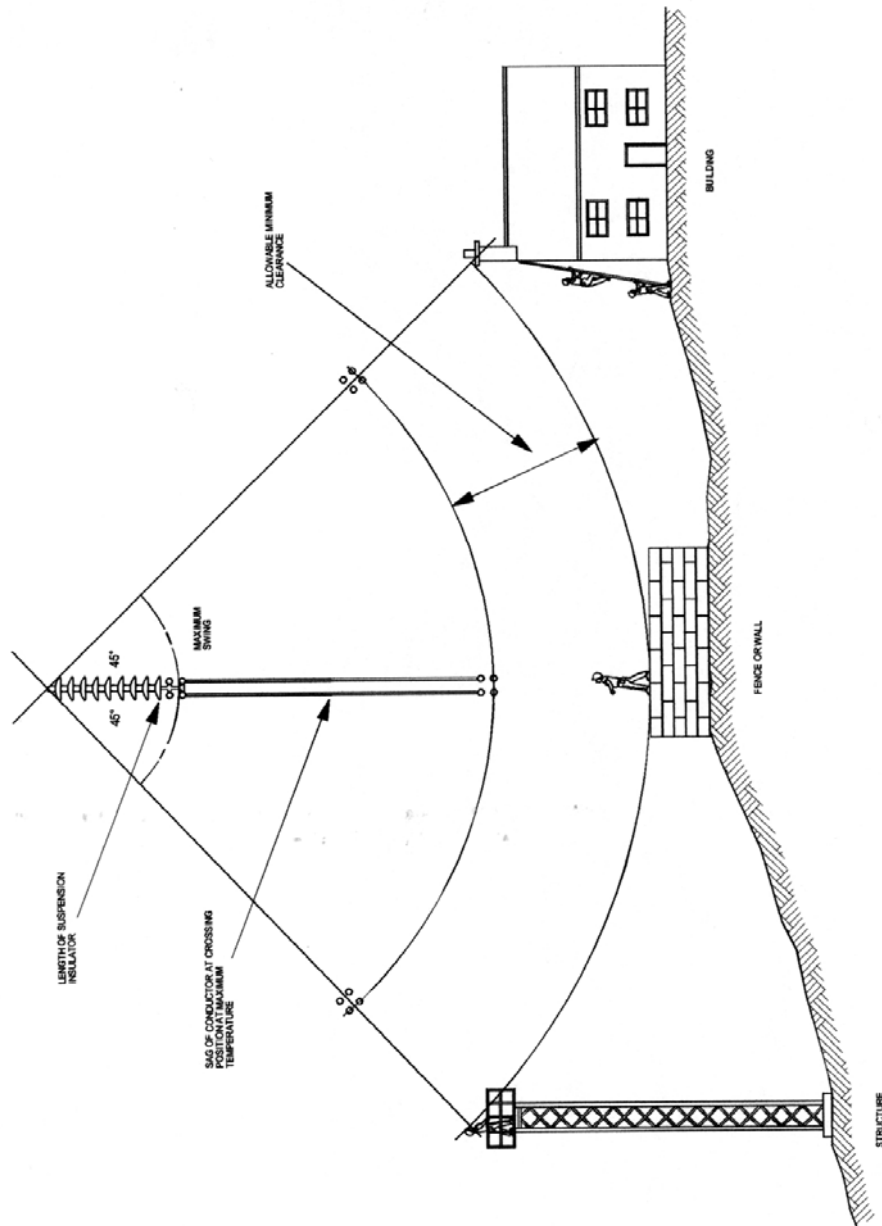
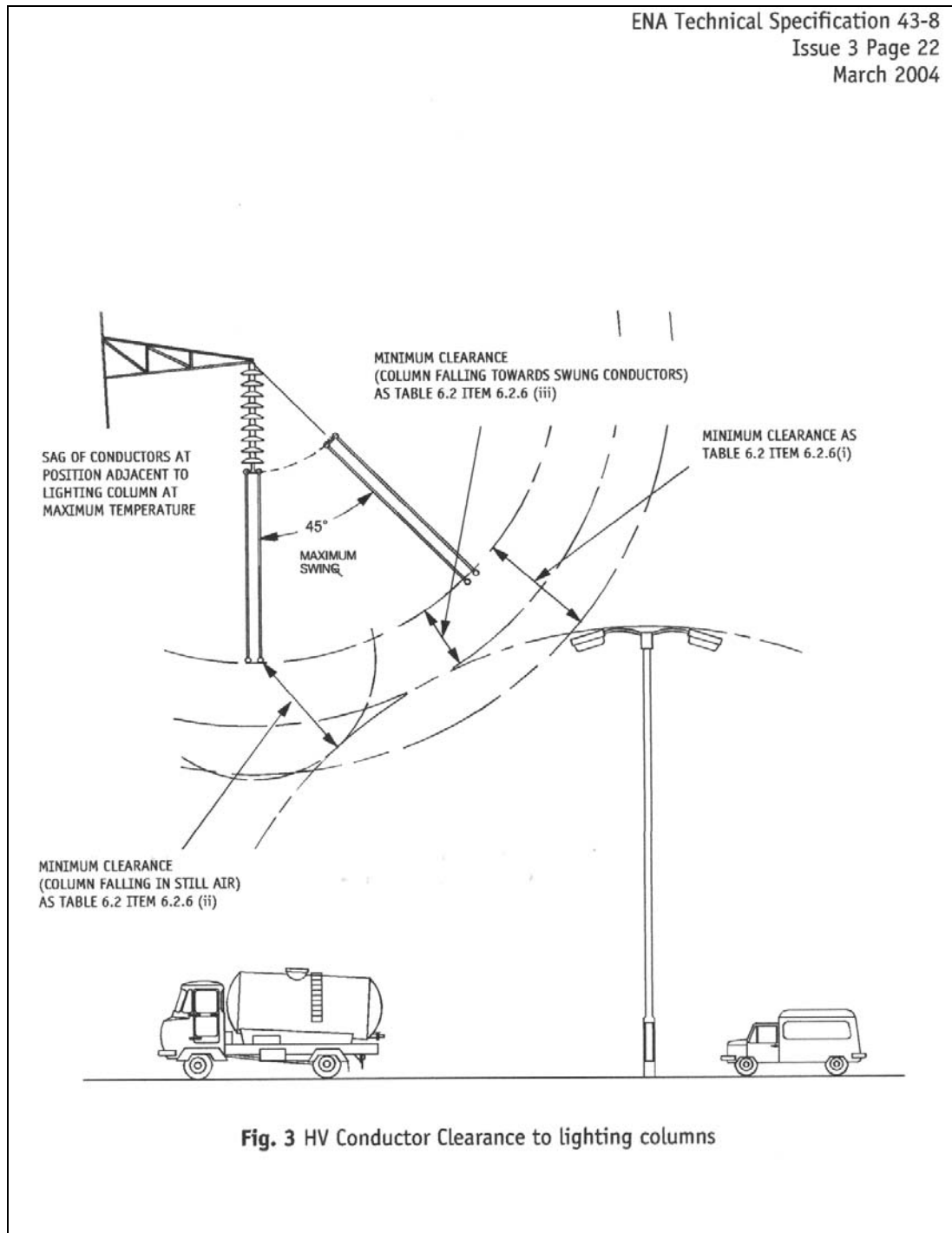


Fig. 1 Clearance to objects (on which a person can stand)

Clearance to Objects (on which a person might stand)
CE Electric UK Drawing Ref – Y101L0201

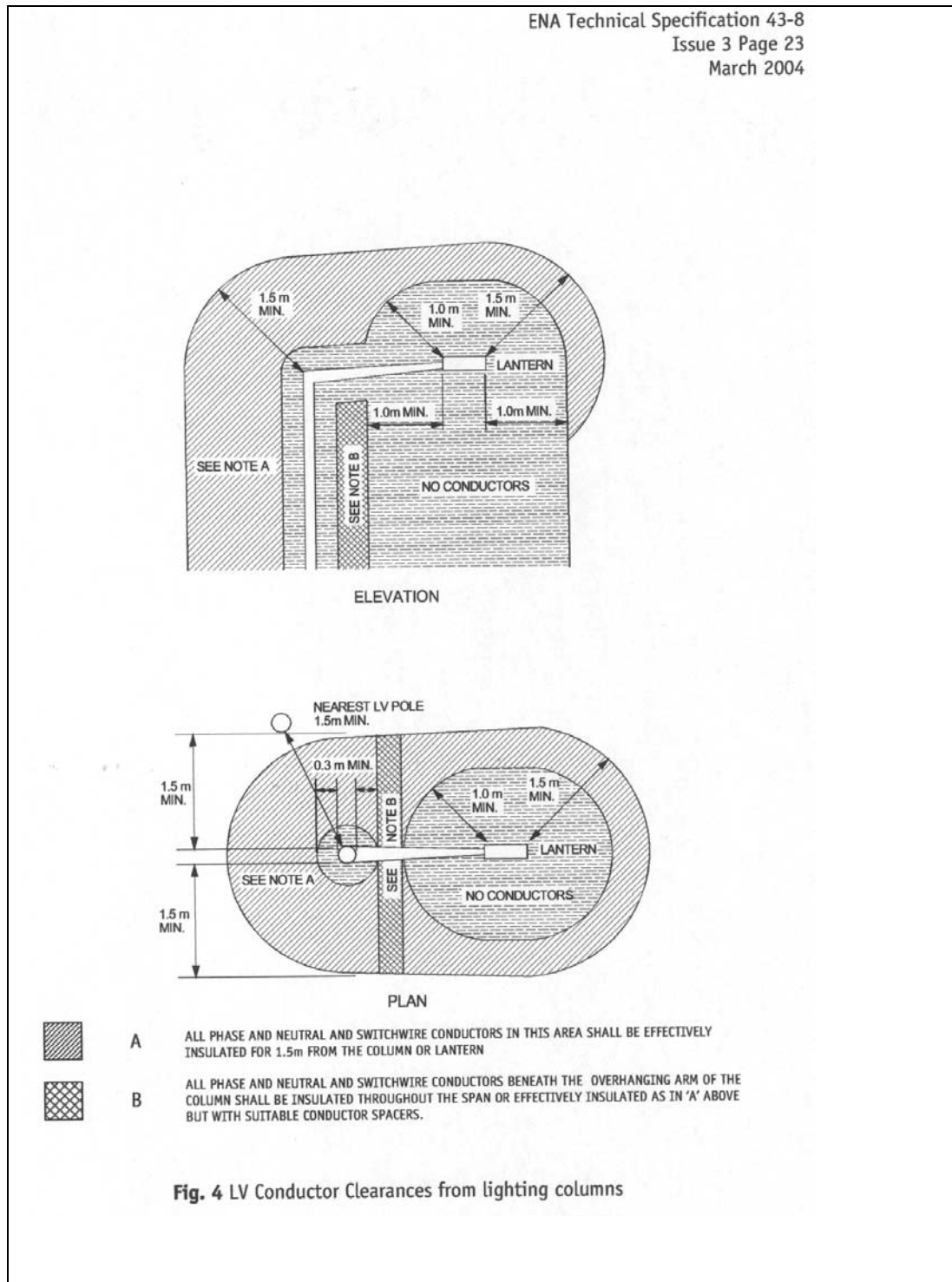


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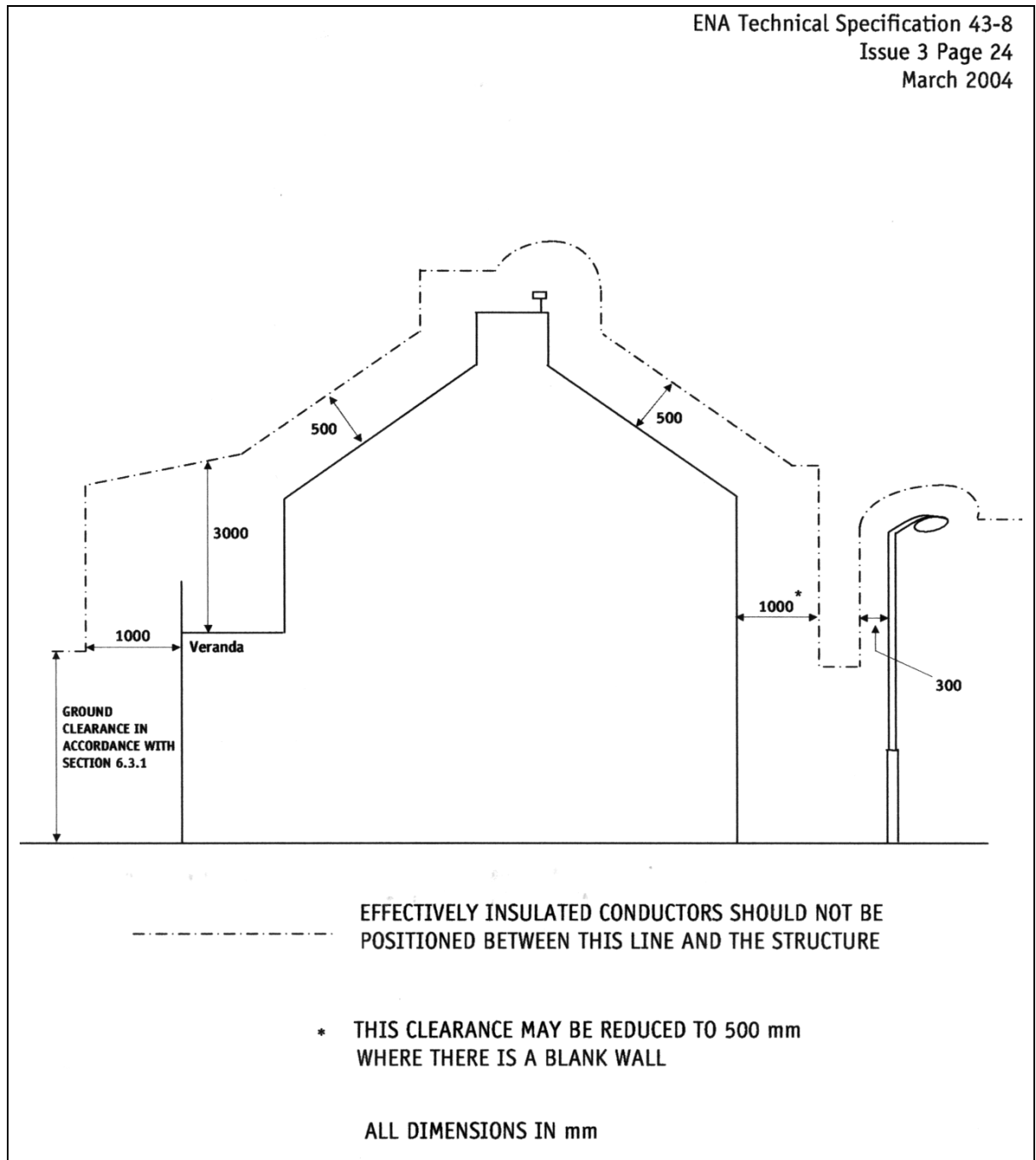
HV Conductor Clearance to lighting columns
CE Electric UK Drawing Ref – Y101L0203

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LV Conductor Clearances from lighting columns
CE Electric UK Drawing Ref – Y101L0204

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**Clearance between structures and effectively insulated
conductors installed on poles
CE Electric UK Drawing Ref – Y101L0205**